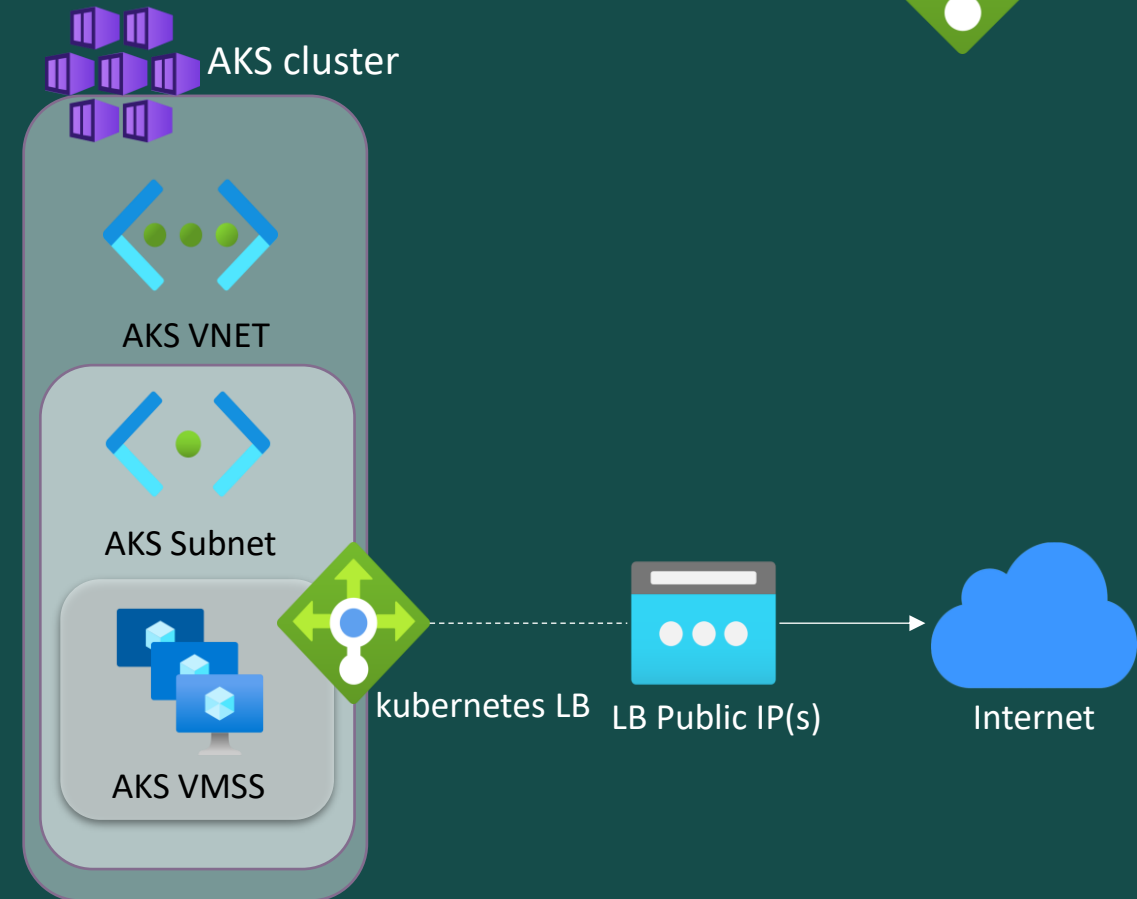


Outbound types in AKS - Load Balancer



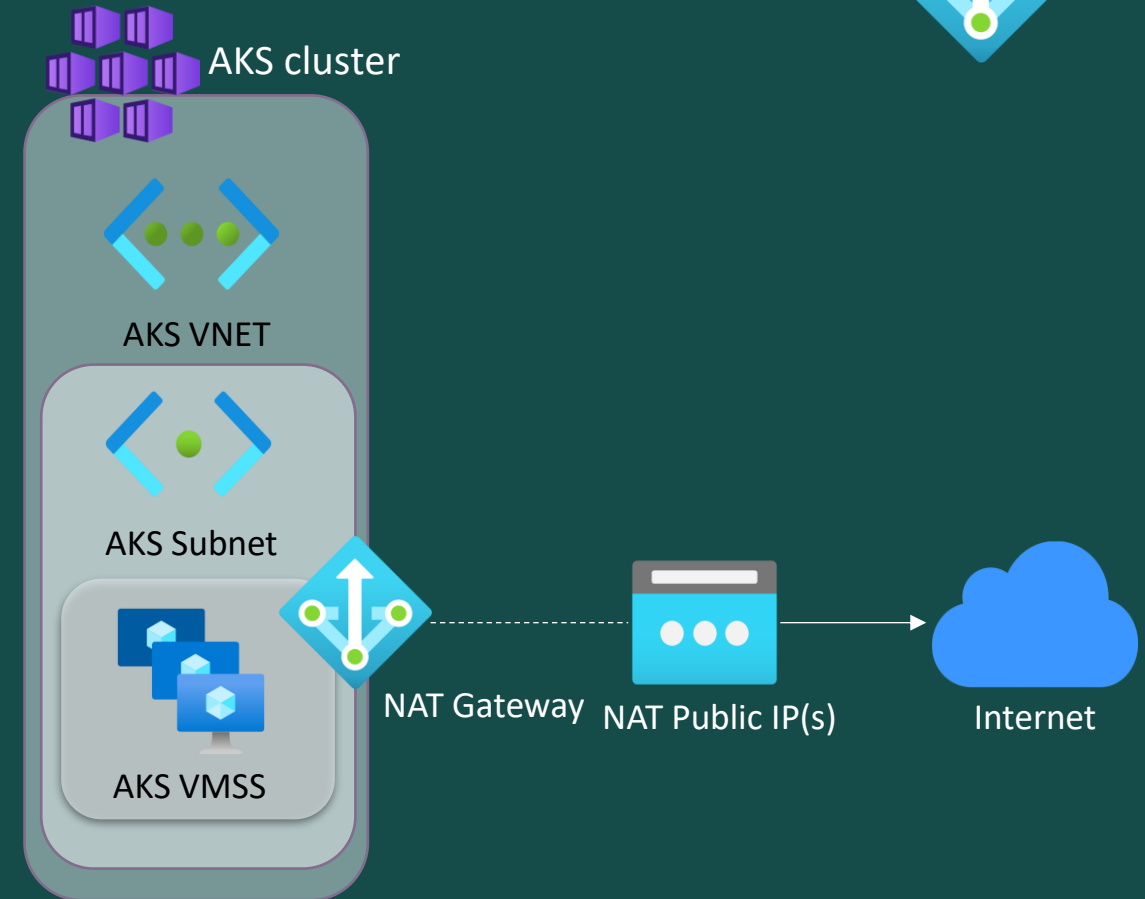
- Functionalities in AKS:
 1. **Provides outbound connectivity to nodes**
 2. Exposes applications via Kubernetes service type LB
- Default outbound type in AKS
- Node's private IP is translated using SNAT to LB's frontend IP using SNAT to allow outbound connectivity
- SNAT ports are allocated per node
- You can configure multiple public IPs for the LB
 - `--load-balancer-managed-outbound-ip-count`
 - `--load-balancer-outbound-ips`
- You can set the outbound ports per node with: `--load-balancer-outbound-ports`
- You can set the idle timeout with: `--load-balancer-idle-timeout`



Outbound types in AKS – NAT Gateway



- Assigned at subnet level. Can be allocated to multiple subnets, but one subnet cannot use more than one NAT Gateway
- For NAT gateway the ports are consumable by all VMs within the subnet(s), comparing with LB where they are assigned per node
- If using outbound type NAT Gateway, an Azure Load Balancer will be created only if a Kubernetes service type Public Load Balancer will be created
- `--outbound-type managedNATGateway`
- `--outbound-type userAssignedNATGateway --vnet-subnet-id`
- `--nat-gateway-idle-timeout`
- `--nat-gateway-managed-outbound-ip-count {up to 16}`
- After cluster creation, you can also change the outbound type between load balancer, managed NAT gateway, and user-defined NAT gateway



Outbound types in AKS - UDR



- With UDR, AKS won't build up egress paths automatically. You are required to configure the egress
- AKS must be deployed into an existing VNET/subnet configured to explicitly route egress traffic
- Azure Firewall can be used as a NVA to control outbound traffic, but there are specific requirements
- A Load Balancer will be created only after creating a Kubernetes service type Public Load Balancer
- Azure Firewall provides 2496 ports per public IP; up to 250 public IPs can be assigned

